



EC400

Lecture 1

Foundations

Probability and Statistical Inference ■ Tom Glinnan

Aim of the Course

- This course covers all the stats background you need for graduate courses, especially in econometrics
- The big aim is **intuition**.
- You can probably do a lot of the math already; the aim is for you to really understand what you're doing and why

Lecture Topics

1. Foundations
2. Statistics I - Basics, Regression
3. Statistics II - Technical details, connection to linear algebra
4. Hypothesis Testing
5. OLS and Asymptotics

Full syllabus is up on moodle.

About me

- Assistant Professor (Education) in economics department
- Email: t.m.glinnan@lse.ac.uk (please no explanatory questions over email)
- PhD from LSE, research area is econometric theory.
- Have taught across most stats / metrics courses from UG - PhD. This year includes PhD Economics seminars.

- 5 days. Each day: 2.5h lecture, 2.5h seminar.
- All lectures are recorded and available on moodle.
- Problem sets covered in the seminar. Please look before if time (but nothing to hand in). There are also unseen questions.
- **No required textbook** - if you actively want one you can see the list in the syllabus on moodle.

Exam

- Exam is on Tuesday 22nd in the morning. Remedial sessions if that doesn't go well in AT.
- No daily OH for the course, so please make sure to ask seminar teachers in the seminar.
- Examinable content = what I talk about in lectures + any concepts explicitly introduced in the seminar. **Nothing else.**
- Main aim is to diagnose how much is needed for you to succeed on MSc/MRes.
- You'll have 2 sample exams; also past exams and solutions. However: somewhat different syllabus.

Prerequisites

- High school math, multivariate calculus...
- The main thing we don't have time to talk about is linear algebra.
- See guide on moodle (pdf), and also series of short videos.
- **Not directly tested** in this course, but we do build on some of the concepts. Also directly relevant for graduate courses.

Any questions?

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

What are we trying to do?

- Simply describe the data ('How correlated is x with y ?')
- Estimate causality ('What's the causal effect of x on y ?')
- Forecasting ('How can I predict y_j from some new x_j ?')
- ...

The Basics

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

Two parts:

- What does β mean?
- Given a β , how can we construct a good guess $\hat{\beta}$?

You may have only ever seen Step 2. Real work involves **both**.

A Two-Step Process

Identification

If we had infinite amounts of data, could we learn what we want to learn?

Interpreting β
as something we care about

Estimation

How do we use the sample to guess it?

Estimating β using some $\hat{\beta}$

- You nearly always need **assumptions** to be able to do both things.
- A clean way to see the separation is with an RCT.

Plan for today

- An RCT: population targets, identification and estimation.
- Sample statistics, quantiles and probability.
- Frequentist and Bayesian interpretations of probability.
- Estimators, estimates and repeated-sampling distributions.
- Unbiasedness, efficiency, consistency and mean squared error.

Today: the big picture. From tomorrow: the formal tools.

The RCT example

- Banerjee, Cole, Duflo and Linden (2007) study remedial education in India.
- The *Balsakhi* program used local tutors to help children falling behind in basic literacy and numeracy.
- Some school-grade cells were assigned to receive the tutor; others were assigned to comparison.
- Outcomes were measured using test scores.

Source for details: Banerjee, Cole, Duflo and Linden (2007); J-PAL Balsakhi evaluation summary.

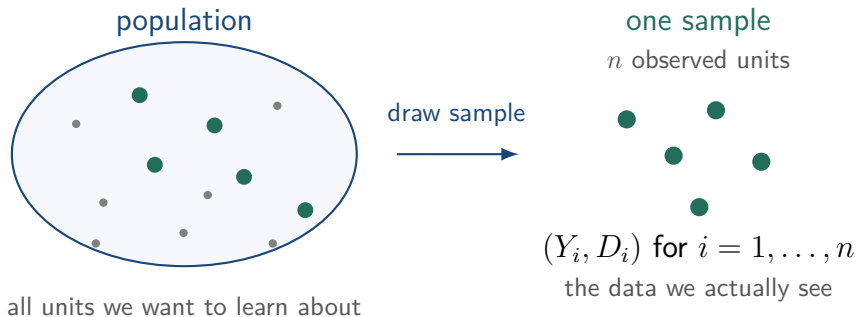
The question

Did the tutoring program improve learning?

The Plan:

1. **Target:** What is the thing we want to estimate?
2. **Identification:** under some assumptions, this is the β in some regression.
3. **Estimation:** under some assumptions, a guess $\hat{\beta}$ is likely close to β .

What do we observe?



Step 0: The target

‘Did the program improve outcomes’ is a **causal** question

- $Y_i(1)$: outcome for person i if they were assigned the tutoring program.
- $Y_i(0)$: outcome for person i if they were not assigned the tutoring program.
- These are hypothetical.

Imagine taking person i and moving them to a world where their treatment status changed, keeping everyone else the same.

Potential outcomes

- Each unit has two possible outcomes:

$$Y_i(1), \quad Y_i(0).$$

- We observe the one corresponding to the assigned status:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

Unit	D_i	$Y_i(1)$	$Y_i(0)$
A	1	observed	missing
B	0	missing	observed
C	1	observed	missing
D	0	missing	observed

So $D_i = 0$ implies $Y_i = Y_i(0)$ and $D_i = 1$ implies $Y_i = Y_i(1)$

Treatment effects

- We define the treatment effect of the program as:

$$\tau_i := Y_i(1) - Y_i(0).$$

- **Why?** the only thing different in the two worlds is whether or not you were treated.
- So any difference in outcomes is because you were treated.
- **Note:** different τ_i for every i . τ_i fundamentally unknowable.

Objects of interest

- Average treatment effect:

$$ATE = \mathbb{E}[Y(1) - Y(0)].$$

- Average treatment effect on the treated:

$$ATET = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

- Quantile TEs, TEs for certain subgroups, etc, etc.

In our example:

Our ultimate target is the **ATE**.

Step 1: Identification

Let

$$\beta := \mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0).$$

If we knew the **population** (instead of just the sample), we could calculate this.

Note: the same as the β from:

$$Y_i = \alpha + \beta D_i + \varepsilon_i \quad \mathbb{E}(\varepsilon_i \mid D_i) = 0$$

(since $\mathbb{E}(Y_i \mid D_i = 0) = \alpha$ and $\mathbb{E}(Y_i \mid D_i = 1) = \alpha + \beta$)

Idea: under some assumptions, this is equal to the ATE.

β and the ATET

Start with

$$\begin{aligned}\beta &= \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] \\ &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0].\end{aligned}$$

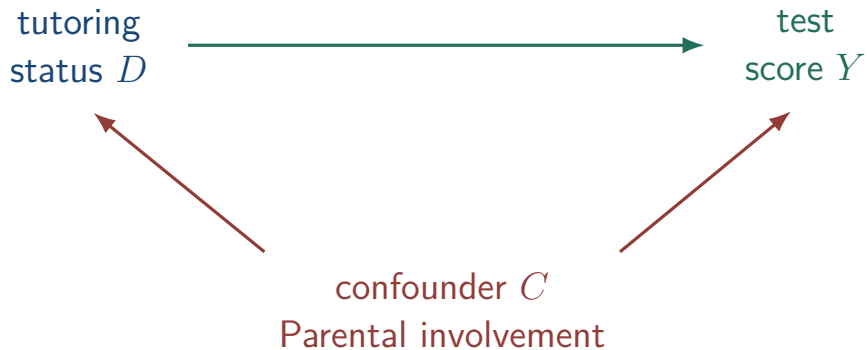
Add and subtract $\mathbb{E}[Y(0) \mid D = 1]$:

$$\begin{aligned}\beta &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1] \\ &\quad + \left[\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \right] \\ &= \text{ATET} + \underbrace{\left[\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \right]}_{\text{Selection Bias}}.\end{aligned}$$

Selection Bias

- Suppose parents could sign their children up to the sessions
- More educationally involved parents $\implies$ higher $Y(0)$
- Therefore $\mathbb{E}[Y(0) \mid D = 1] > \mathbb{E}[Y(0) \mid D = 0]$, so $\beta > ATET$
- Possible to have *negative* ATET and still see positive β
- **Correlation versus causation** is about whether $\beta = ATE$

A confounder



No selection bias

Random assignment implies

$$D \perp\!\!\!\perp (Y(0), Y(1)).$$

In particular,

$$\mathbb{E}[Y(0) \mid D = 1] = \mathbb{E}[Y(0) \mid D = 0].$$

- The assigned tutoring group is not systematically different in how it would have done without tutoring.
- The comparison group can stand in for the treated group's missing counterfactual.

As such, random assignment implies $\beta = \text{ATET}$

$$\beta = \text{ATE}$$

Furthermore, random assignment makes $\text{ATET} = \text{ATE}$.

$$\begin{aligned}\text{ATET} &:= \mathbb{E}(Y(1) - Y(0) \mid D = 1) \\ &= \mathbb{E}(Y(1) \mid D = 1) - \mathbb{E}(Y(0) \mid D = 1) \\ &= \mathbb{E}(Y(1)) - \mathbb{E}(Y(0)) = \text{ATE}\end{aligned}$$

In summary: randomizing $D \implies$ random assignment holds $\implies \beta = \text{ATE}$

Step 2: Estimation

- In the sample, we observe $(Y_i, D_i)_{i=1}^n$
- We know that we want to estimate $\beta := \mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0)$
- A natural statistic is the difference in average outcomes:

$$\hat{\beta} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i.$$

Note: same as OLS $\hat{\beta}$ in the equation

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

Toy numbers

School-grade cell	D_i	Endline score Y_i	Group
A	1	64	tutoring
B	1	72	tutoring
C	1	68	tutoring
D	0	55	comparison
E	0	61	comparison
F	0	60	comparison

$$\hat{\beta} = \frac{64 + 72 + 68}{3} - \frac{55 + 61 + 60}{3} = 9.33.$$

Did the tutoring program improve learning?

The Solution:

1. **Target:** We want to estimate the ATE of the program
2. **Identification:** By making the program an RCT, we have random assignment, therefore $\beta = \text{ATE}$
3. **Estimation:** We can estimate β by comparing the means of treatment and control groups in our sample

Aside: this is the formal theory of the **scientific method**.

A cautionary tale

What happens if we don't show identification first?

e.g. we just do OLS to get $\hat{\beta}$ from

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

Then

$$\hat{\beta} \approx \beta = \text{ATET} + \text{Selection Bias} \neq \text{ATE!}$$

No amount of data can fix bad identification

Identification beyond causality

More generally: If I had infinitely many observations of the variables that I see, would I know what I wanted to know?

Forecasting:

- **Target:** what future returns on a stock might be
- Even if you can predict perfectly on existing data, how does that link to the future?

Structural Work:

- **Target:** understanding people's preferences from choices (eg political)
- Are the choices rich enough to tell you?

Causal identification

- Each design tries to justify a comparison group.
- The shared question is: why does this comparison reveal the missing counterfactual?

Matching

Compare treated and untreated units with similar observed characteristics.

Regression discontinuity

Compare units just on either side of a threshold.

Instrumental variables

Use external variation that shifts treatment.

Difference-in-differences

Compare changes over time across groups.

Estimation

Estimation

The rest of the course:

Statistics is about how to choose a $\hat{\beta}$ to estimate a given β , and theoretical properties which formalize **when, why and how** $\hat{\beta} \approx \beta$

Probability theory is the language that this theory is written in

Common sample statistics

A **statistic** is any function of the observed sample:

$$T_n = g(X_1, \dots, X_n).$$

- **Location:** mean, median and quantiles.
- **Spread:** variance, standard deviation and interquartile range.
- **Dependence:** sample covariance and correlation.
- **Shape and tails:** skewness, kurtosis and extreme quantiles.
- Each statistic has a sampling distribution across repeated samples.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i,$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

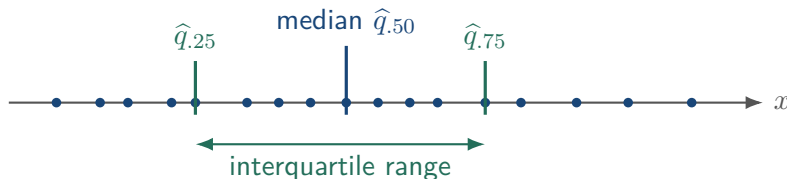
Sample quantiles

Sort the observations to obtain the **order statistics**:

$$X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}.$$

For $0 < p \leq 1$, the sample p -quantile is a value with about a fraction p of the data at or below it. A simple convention uses

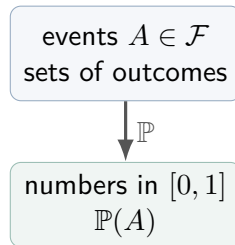
$$\hat{q}_p = X_{(\lceil np \rceil)}.$$



Probability Theory

- The **sample space** Ω contains every possible outcome.
- An **event** A is a set of outcomes, drawn from a collection $\mathcal{F}$ (sigma algebra).
- A probability function assigns a number to each event:

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1], \quad A \longmapsto \mathbb{P}(A).$$



One fair coin flip

Sample space

$$\Omega = \{H, T\}.$$

These are all the outcomes that could occur.

An event

$$A = \{H\}.$$

This is the event “the coin lands heads.”

Probability function

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\{H\}) = \frac{1}{2}, \quad \mathbb{P}(\{T\}) = \frac{1}{2}, \quad \mathbb{P}(\Omega) = 1.$$

Sigma algebra

$$\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\}.$$

This is the collection of events to which we assign probabilities.

The three probability rules

A probability map obeys three basic rules:

- Total mass:

$$\mathbb{P}(\Omega) = 1.$$

- Non-negativity: $\mathbb{P}(A) \geq 0$ for every event A .
- Addition for disjoint events: if the A_j cannot occur together, then

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \mathbb{P}(A_j).$$

These rules imply $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Two interpretations

Bayesian: Probability = Strength of Belief

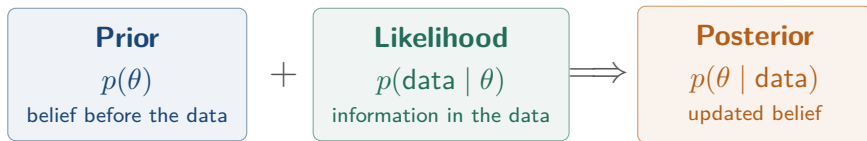
- $\mathbb{P}(\text{heads}) = 1/2$ means 'I'm as equally sure that the coin flip will be heads as it is tails'
- But if I'm a rational person, I should update beliefs according to Bayes rule (how to incorporate new information)

Frequentist: Probability = Long Run Averages

- $\mathbb{P}(\text{heads}) = 1/2$ means 'If I flip a coin many times, about half of them will be heads'
- Difficult to think of eg probability it rains tomorrow.

Bayesian statistics

Bayesian inference updates what we believed before seeing the data.



$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta) p(\theta)}{p(\text{data})}$$

posterior = likelihood $\times$ prior, scaled to integrate to one

That's it! Much focus is in computation.

The running example

Focus in this course: **frequentist** stats.

In our RCT example, our $\hat{\beta}$ was the difference in means

$$\hat{\beta} = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i.$$

and we calculated $\hat{\beta} = 9.33$. This is called our **estimate** for β

The key idea: what would we have guessed if we had seen different data?

Estimator vs estimate

- An **estimator** is a random rule:

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0.$$

- Before data are realized, it can take different values.

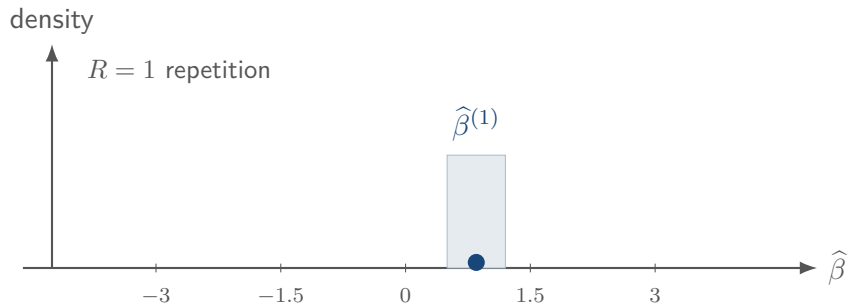
- An **estimate** is the number produced in one realized sample.

- In the toy data:

$$\hat{\beta} = 9.33.$$

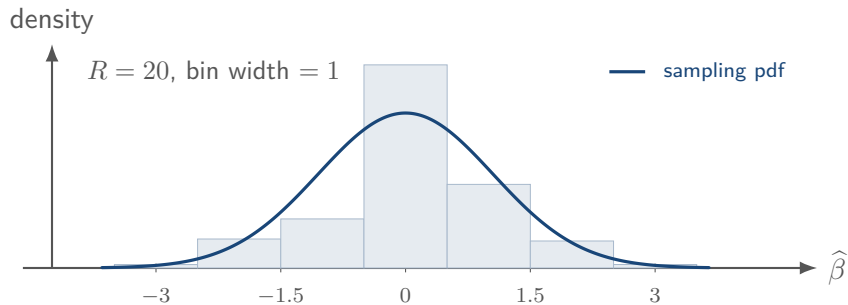
Terminology: we only ever see **one** sample, consisting of n datapoints / observations.

One sample



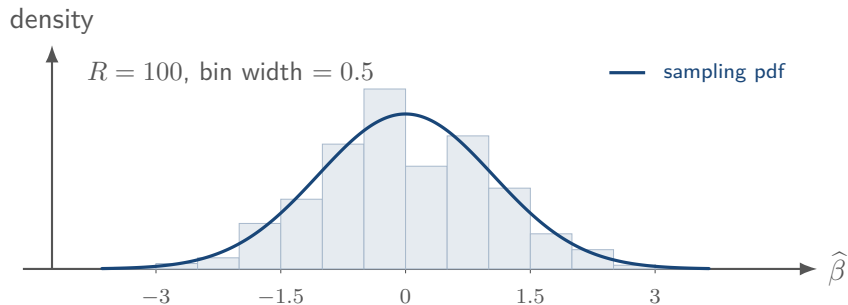
One sample of n observations produces one estimate—not enough to reveal a distribution.

Twenty repeated samples



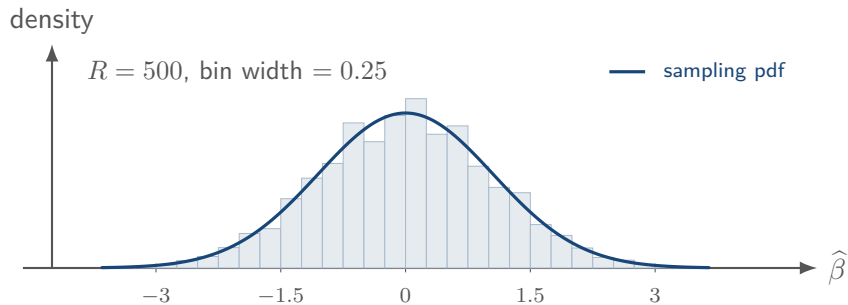
Twenty estimates give only a coarse outline of where the estimator tends to land.

One hundred repeated samples



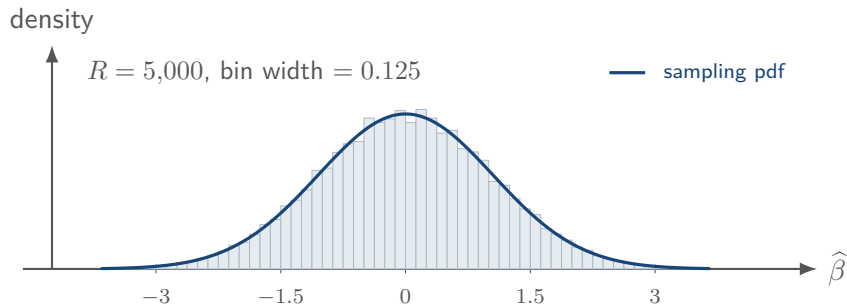
More repetitions support narrower bins; the histogram begins to settle around the pdf.

Five hundred repeated samples



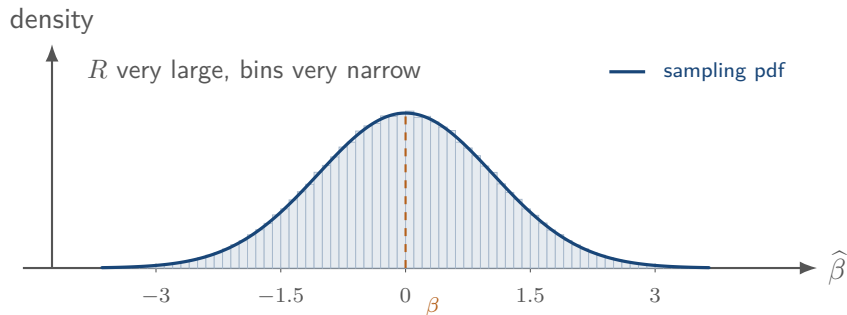
Narrower bars reveal the shape; the remaining bumps are finite-repetition noise.

Five thousand repeated samples



With 5,000 repetitions, the upper edges of the bars closely trace the smooth pdf.

Sampling distribution



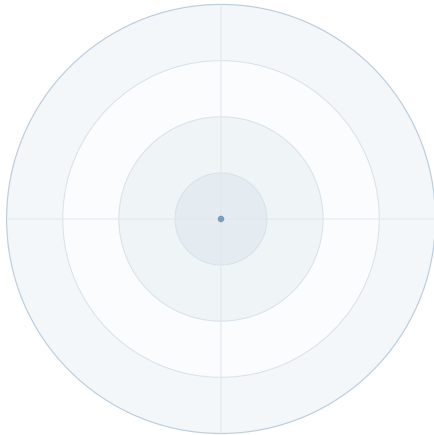
Sampling distribution: the distribution of $\hat{\beta} = g(S)$ across repeated samples of the same size.

As R grows and the bins narrow, the histogram reveals its smooth pdf.

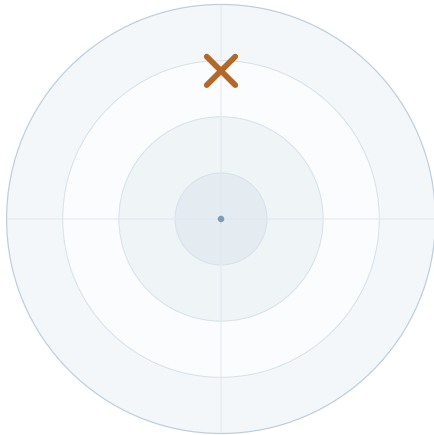
If my DGP satisfies certain assumptions,
then my estimator $\hat{\beta}$ satisfies certain properties.

- Unbiasedness, consistency and efficiency all formalize different aspects of what makes a **good** estimator.
- They're properties on an estimator's distribution.

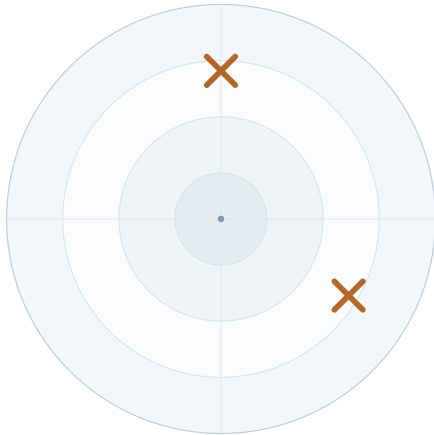
Unbiasedness



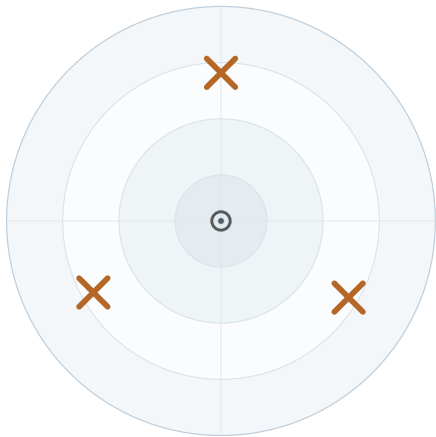
Unbiasedness



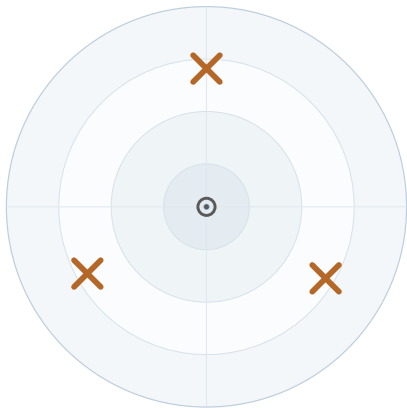
Unbiasedness



Unbiasedness

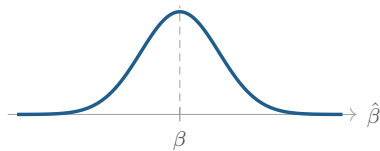


Unbiasedness

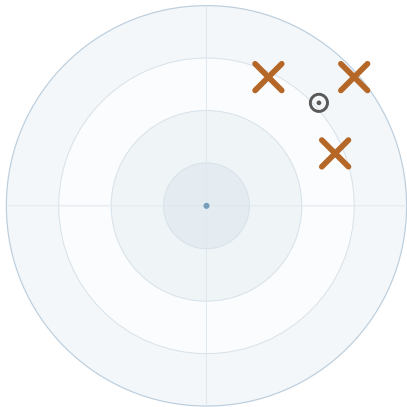


Unbiased

$$\mathbb{E}(\hat{\beta}) = \beta$$

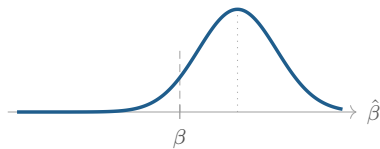


Bias

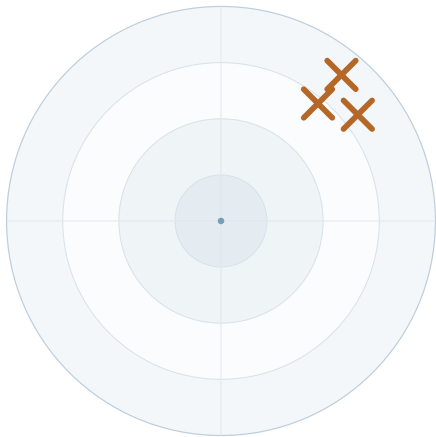


Biased

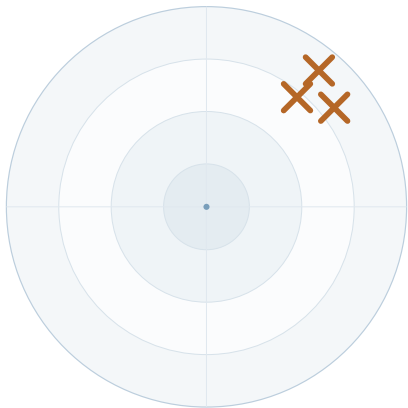
$$\mathbb{E}(\hat{\beta}) \neq \beta$$



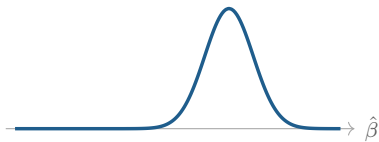
Efficiency



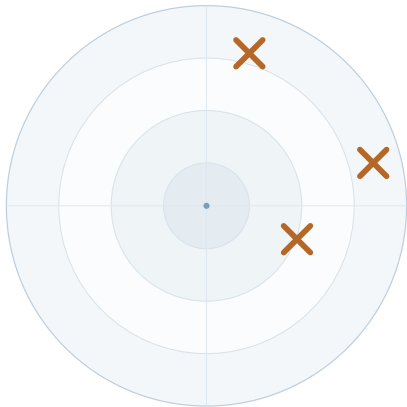
Efficiency



Low variance
 $\text{Var}(\hat{\beta})$ is small

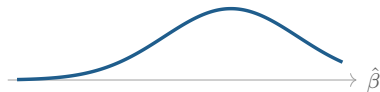


Efficiency



High variance

$\text{Var}(\hat{\beta})$ is large



Efficiency is relative

- “Low variance” needs a target and a class of estimators.
- Efficient usually means lowest variance *within that class*.
- A constant guess has zero variance, but it can be badly wrong.

Gauss–Markov

Best linear unbiased estimator
under the classical linear model.

Cramer–Rao

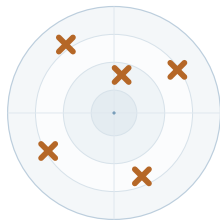
Lower bound for unbiased estimators
in regular parametric models.

More data bring us closer to the target.

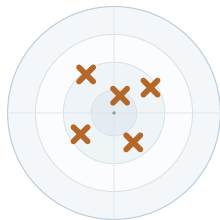
- For each sample size, repeated samples produce a sampling distribution.
- As n grows, that distribution becomes more concentrated around the true β .

We call this consistency.

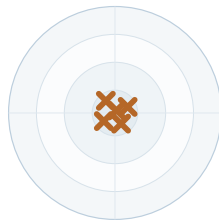
Consistency on the target



$n = 25$



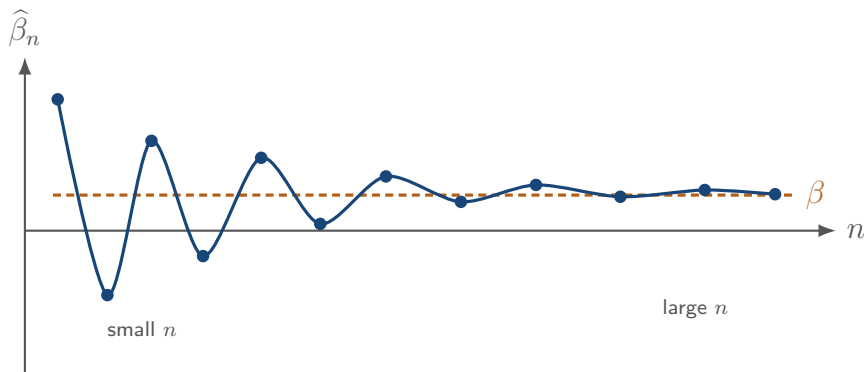
$n = 100$



$n = 1000$

As the sample grows, more of the sampling distribution lies close to the target.

Consistency over n



- Consistency is about what happens as the sample size grows.
- With more data, estimates tend to be closer to the target β .

Defining consistency

1. $|\hat{\beta} - \beta|$ is a measure of distance between $\hat{\beta}$ and β
2. Pick any $\varepsilon > 0$. We'll say that $\hat{\beta}$ is 'far away' from β if the distance $> \varepsilon$
3. For example, if $\varepsilon = 1/2$, then $\hat{\beta}$ is 'far away' from β if the distance $> 1/2$
4. So $\mathbb{P}(|\hat{\beta} - \beta| > \varepsilon)$ is the probability that $\hat{\beta}$ is 'far away' from β .
5. But I could have chosen $\varepsilon = 2$, which would be a different notion of 'far away'

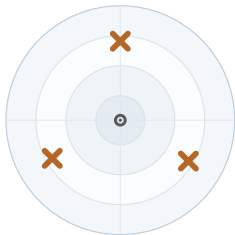
As such,

$$\forall \varepsilon > 0 \quad \mathbb{P}(|\hat{\beta}_n - \beta| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

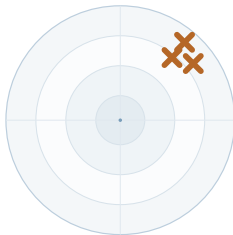
means:

'No matter how I define far away, the probability of being far away goes to 0 as n goes to infinity'

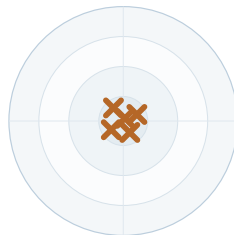
Three estimator properties



Unbiased
 $\mathbb{E}(\hat{\beta}) = \beta$

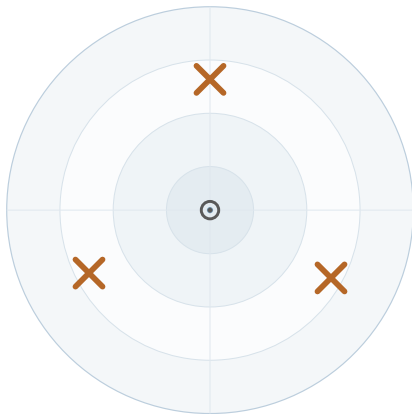


Efficient
low $\text{Var}(\hat{\beta})$



Consistent
 $\hat{\beta}_n \xrightarrow{p} \beta$

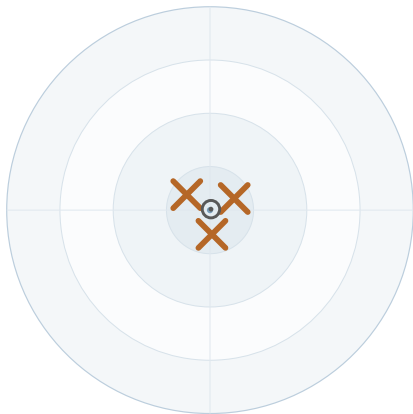
Bias and variance



Low bias

$$\text{Bias}(\hat{\beta}) \approx 0$$

Bias and variance



Low bias + low variance

$$\text{Bias}(\hat{\beta}) \approx 0, \quad \text{Var}(\hat{\beta}) \text{ small}$$

Bias and variance

- Bias is $\text{Bias}(\hat{\beta}) = \mathbb{E}(\hat{\beta}) - \beta$.
- MSE is the repeated-sampling average of squared distance to the truth.

$$\text{MSE}(\hat{\beta}) = \mathbb{E}[(\hat{\beta} - \beta)^2] = \text{Bias}(\hat{\beta})^2 + \text{Var}(\hat{\beta})$$

In shorthand: $\text{MSE} = B^2 + V$.

Low MSE requires bias and variance to be small together.

Large and small samples

Very large samples

Consistency is often the key requirement:
eventually $\hat{\beta}_n$ concentrates near β .

Small samples

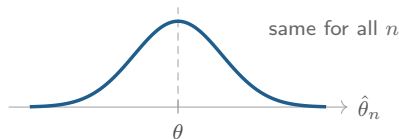
Consistency is only an asymptotic
promise:
bias and variance may matter more.

Unbiasedness without consistency

Toy model: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1)$.

- Estimator: $\hat{\theta}_n = X_1$.
- Unbiased: $\mathbb{E}(\hat{\theta}_n) = \theta$.
- But $\hat{\theta}_n \sim N(\theta, 1)$ for every n .
- The spread never shrinks, so $\hat{\theta}_n$ does not concentrate around θ .

Unbiased, but not consistent

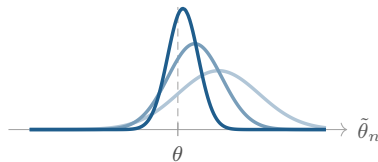


Consistent but biased

Same toy model: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta, 1)$.

- Start with the sample mean $\bar{X}_n$, then define $\tilde{\theta}_n = \bar{X}_n + 1/n$.
- Finite-sample bias: $\text{Bias}(\tilde{\theta}_n) = 1/n \neq 0$.
- Distribution: $\tilde{\theta}_n \sim N(\theta + 1/n, 1/n)$.
- The bias vanishes and the spread shrinks, so $\tilde{\theta}_n \xrightarrow{p} \theta$.

Consistent, but biased



Estimator conditions

Estimator	Unbiased?	Consistent if...
OLS	Yes: classical model.	Exogeneity; enough variation.
IV / 2SLS	Usually biased .	Valid, relevant instruments.
MLE	Often biased.	Correct likelihood.
GMM	Often biased.	Valid moments.

Takeaway

- Unbiasedness: the repeated-sampling average hits the target.
- Efficiency: the sampling distribution is tight, relative to a class.
- Consistency: the estimator moves toward the target as the sample grows.

Good finite-sample performance is about bias and variance.

Good large-sample performance is about the probability limit.

Two links

Statement	Needed assumptions
$\hat{\beta}$ estimates β	Sampling and probability assumptions.
β equals the ATE	Identification assumptions.
$\hat{\beta}$ estimates the ATE	Both sets of assumptions.

Estimation **cannot** repair a broken identification argument.

Starting with $\hat{\beta}$

An alternative way of looking at things is to view the steps the other way round, by starting with the estimator.

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0.$$

- Under sampling assumptions,

$$\hat{\beta} \xrightarrow{p} \mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0).$$

- That is the population difference in observed outcomes.
- Identification asks whether this limit is also the causal target.

This way of viewing things is common in eg applied econometrics.

Key messages

- Real data work needs two steps: Identification and Estimation
- Identification is only trivial if you aim is to literally describe data. Otherwise not!
- Stats can be Frequentist or Bayesian. Unbiasedness, Efficiency and Consistency are frequentist ideas
- They formalize what makes a good estimator: ie when we might $\hat{\beta} \approx \beta$
- Much of statistics / econometrics is about proving if a DGP has certain assumptions, then my estimators have these properties

Next time

Lecture 2: Statistics I